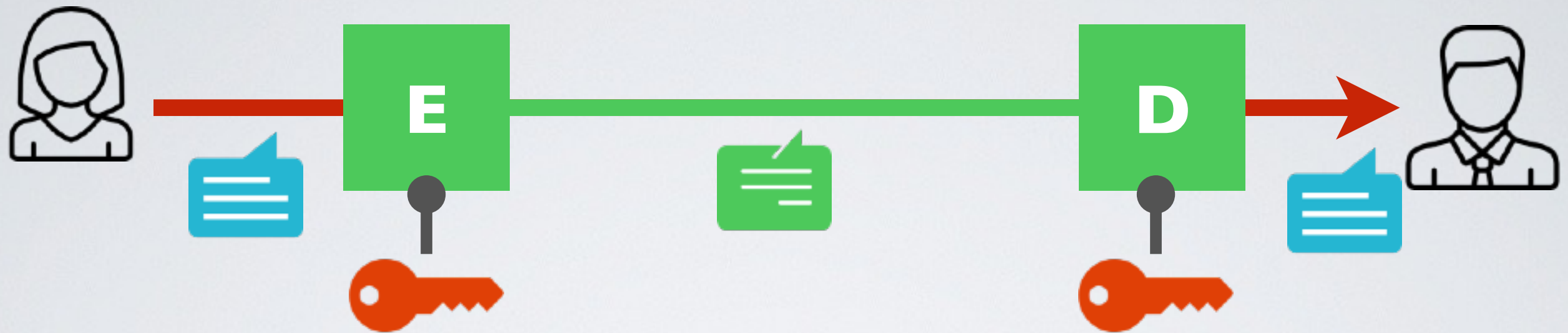


Functional Requirements



➡ The same key k is used for encryption E and decryption D

1. $D_k(E_k(m))=m$ for every k , E_k is an injection with inverse D_k
2. $E_k(m)$ is easy to compute (either polynomial or linear)
3. $D_k(c)$ is easy to compute (either polynomial or linear)
4. $c = E_k(m)$ finding m is hard without k (exponential)

Two Families of Symmetric Encryption Schemes

Stream cipher

RC4 (now deprecated)

ChaCha20

Block cipher

- Encryption standards

DES (and 3DES) - Data Encryption Standard (now deprecated)

AES - Advanced Encryption Standard

- Block cipher modes of operation